

**PSG Institute of Technology and Applied Research, Coimbatore - 641 062**

**Internal Assessment I**

**AD3351 - DESIGN ANALYSIS AND ALGORITHMS**

**II Year B. Tech (AI&DS) - Third Semester**

**Assignment 1**

**Asymptotic Analysis & Forward and Backward Substitution (CO2)**  
**(10 Marks)**

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### **Asymptotic Notation Properties**

1. Prove that if  $f(n) = O(g(n))$  and  $g(n) = O(h(n))$ , then  $f(n) = O(h(n))$ .
  2. Show that  $3n^2 + 5n + 2 = \Theta(n^2)$ .
  3. Verify whether  $2^{n+1} = O(2^n)$  and  $2^{2n} = O(2^n)$ . Justify your answer.
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### **Forward and Backward Substitution**

5. Solve the recurrence relation using forward substitution:  
$$T(n) = T(n-1) + n, T(1) = 1$$
6. Solve the recurrence relation using backward substitution:  
$$T(n) = T(n/2) + 1, T(1) = 1$$
7. Use both forward and backward substitution to solve:  
$$T(n) = T(n-1) + 2n, T(1) = 1$$
 Compare both methods.

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**Assignment 2 :**

**Recurrences Relation using Master Theorem and Tree Method (CO2)**

**(10 Marks)**

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### **Master Theorem**

1. Apply Master Theorem to solve:  $T(n) = 2T(n/2) + n$
  2. Solve using Master Theorem:  $T(n) = 3T(n/2) + n^2$
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### **Tree Method**

3. Solve the recurrence using recursion tree method:  $T(n) = T(n/2) + n$
  4. Use the recursion tree method to solve:  $T(n) = 2T(n/2) + n \log n$
  5. Solve using tree method and verify with Master Theorem:  $T(n) = 4T(n/2) + n^2$
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### **Substitution Method (Advanced Recurrences)**

6. Solve using substitution method:  $T(0)=1, T(1)=3, T(2)=9,$   
 $T(n) = T(n-1) + 2T(n-2) + 3T(n-3), n \geq 3$
  7. Solve using substitution method:  $T(n) = 2T(n-1) + 1, T(0)=1$
  8. Solve using substitution method:  $T(n) = 4T(n-1) - 4T(n-2), T(0)=1, T(1)=4$
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